

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

☎ (587) 879-4626 | ✉ tkuipers123@gmail.com | 🏠 tkuipers.ca | 📱 [tkuipers](#) | 🌐 [tyler-kuipers](#)

About Me

I've spent 10 years writing software, with the last 3 focused on technical leadership while staying hands-on. I still write code daily: building developer tooling, debugging production issues, and shipping features alongside my team. I built a developer platform that cut environment provisioning from weeks to hours and is now used org-wide. I run production systems on Kubernetes and AWS with 99.9% uptime, using Datadog and Elasticsearch for observability. I mentor engineers through pairing, code reviews, and design feedback. My stack is TypeScript, Python, and Java on AWS. Outside work, I run a homelab on Kubernetes and build side projects including a mutation testing framework.

Work Experience

Alberta Motor Association

Edmonton, Alberta

SOFTWARE ARCHITECT

Jan 2023 - Present

- Built self-service developer platform that cut environment provisioning from weeks to hours (60x improvement). Features: environment create/destroy, database snapshots, pull-from-prod. Adopted by 15 engineers across 3 teams, created 120+ environments. (Python, Terraform, AWS, Docker)
- Led modernization of document generation from legacy C# to containerized microservices on EKS. Led 2-3 engineers. Added Datadog observability: failure rates, latency, throughput metrics. Significantly improved reliability over legacy system. (Kubernetes, EKS, Docker, Datadog, Helm)
- Architected event-driven system processing 2.8M events across multiple AWS accounts with 99.9% uptime. Diagnosed and resolved production issues across distributed services. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, TypeScript, Terraform)
- Own 4 REST APIs serving 5 internal teams and external partners. Design API contracts, maintain OpenAPI specs, run code reviews. (Python, TypeScript, REST, OpenAPI)
- Mentor engineers on system design, debugging, and best practices. Author RFCs, lead architecture decisions, member of Architecture Steering Committee.

Arterys Inc. (Acquired by Tempus Labs)

Calgary, Alberta (Remote)

SOFTWARE DEVELOPER

Aug 2021 - Aug 2022

- Built and maintained backend services for SaaS medical imaging platform. Focus on low latency and reliability. (Node.js, TypeScript, AWS)
- Diagnosed and resolved performance bottlenecks using Redis caching and AWS infrastructure tuning.
- Monitored production via Elasticsearch and Kibana. Caught and fixed issues before they hit customers.
- Code reviews and testing for team contributions.

Alberta Motor Association

Edmonton, Alberta

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework from scratch that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later. (Java/Gosu, JUnit)
- Led 5-6 person team using TDD. Delivered on time with no critical defects post-launch. Code reviews, mentored junior developers, collaborated with product stakeholders.
- Led Git migration for 12-15 developers. Created Docker image for local dev environments.

Pleasant Solutions

Edmonton, Alberta

SOFTWARE DEVELOPER

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

Halifax, Nova Scotia

SOFTWARE ENGINEER ASSOCIATE

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

Edmonton, Alberta

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

May 2014 - Sep 2016

- Built web application for spatial/temporal data visualization. (JavaScript, PostgreSQL, PHP)

Education

The Open University
MSC. COMPUTING (SOFTWARE ENGINEERING)
Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)
May 2024 - Ongoing

The King's University
B.S. IN COMPUTER SCIENCE

Edmonton, Alberta
Sep 2012 - Dec 2016

Skills

Developer Tooling & Velocity	Build tooling that multiplies team effectiveness. Created developer platform (60x faster provisioning), test frameworks (30% defect reduction), and reusable libraries adopted org-wide. Focus on removing friction so engineers can ship faster.
Production Debugging & Observability	Diagnose complex issues across distributed systems. Set up and use Datadog, Elasticsearch, Kibana for metrics, logging, and tracing. Find root causes and deliver lasting fixes.
Scalable Systems	Build distributed systems on AWS with 99.9% uptime. Microservices on Kubernetes (EKS), event-driven pipelines, multi-account architectures. Focus on reliability, performance, and maintainability.
Technical Leadership & Mentoring	Lead architecture decisions while staying hands-on. Write design docs (RFCs), run code reviews, mentor engineers through pairing and design feedback. Lift others by sharing context and unblocking problems.

Technologies

Languages	TypeScript, Python, Java, C#, Bash
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS), Kubernetes, Docker, Terraform, Helm
Observability	Datadog, Elasticsearch, Kibana, metrics, logging, tracing
Databases	PostgreSQL, Aurora/RDS, Redis
Architecture	Distributed Systems, Event-Driven, Microservices, REST APIs